GCSE PHYSICS

Higher Tier – Paper 1 style

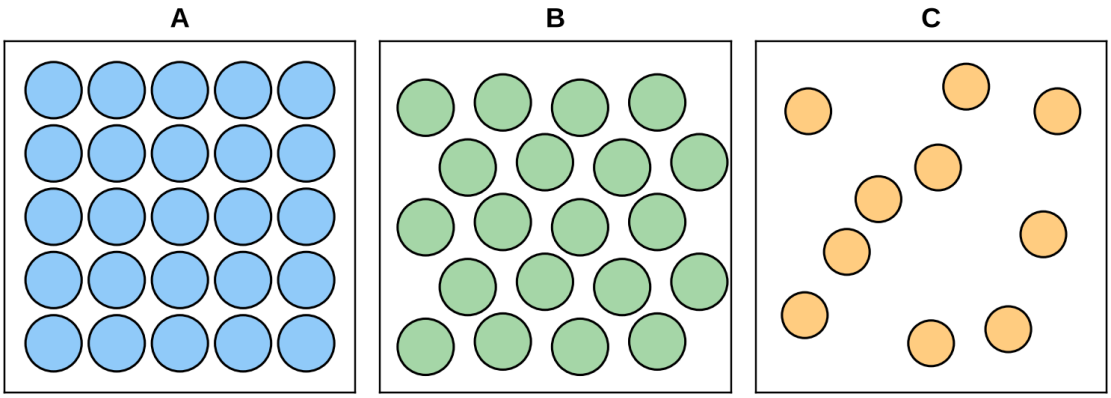
Topic 4.3: Particle model of matter

Practice questions in the style of AQA GCSE Physics (8463). Every question is original; the layout, command words, mark allocations and mark scheme follow the real exam format.

Time allowed	About 150 minutes (roughly 1 minute per mark)
Materials	For this paper you must have: a ruler, a pencil and a calculator.
Instructions	• Use black ink or a black ball-point pen (pencil for diagrams and graphs). • Answer **all** questions. • Answer the questions in the spaces provided. • Do all rough work on a separate sheet. • Cross through any work you do not want to be marked.
Information	• The maximum mark for this paper is **145**. • The marks for questions are shown in brackets. • You are allowed to use a calculator. • You are reminded of the need for good English and clear presentation in your answers. • Questions that ask you to *describe* or *explain* are marked for the science you include, so give each point its own sentence.

Name: _____ Date: _____ Mark: _____ / 145

0 1 The diagrams show the arrangement of particles in a solid, a liquid and a gas.



0 1 . 1 Identify the state of matter shown in each diagram. **[3 marks]**

0 1 . 2 Describe how the particles move in a **solid**. **[2 marks]**

0 1 . 3 Explain why a gas can be compressed easily but a solid cannot. **[3 marks]**

0 1 . 4 Explain, using the diagrams, why a gas has a much lower density than a liquid. **[3 marks]**

0 1 . 5 State what happens to the **mass** of a substance when it changes state. **[1 mark]**

0 1 . 6 A change of state is a **physical** change, not a chemical change. Explain why. **[2 marks]**

0 2 This question is about density.

0 2 . 1 Write down the equation that links density, mass and volume. **[1 mark]**

0 2 . 2 A rectangular aluminium block measures 4.0 cm × 3.0 cm × 2.0 cm. Its mass is 64.8 g. Calculate the density of aluminium in g/cm^3. **[4 marks]**

Answer = _____ g/cm^3

0 2 . 3 Convert your answer to kg/m^3. **[2 marks]**

Answer = _____ kg/m^3

0 2 . 4 The density of water is 1000 kg/m^3. Calculate the mass of 0.50 m^3 of water. **[3 marks]**

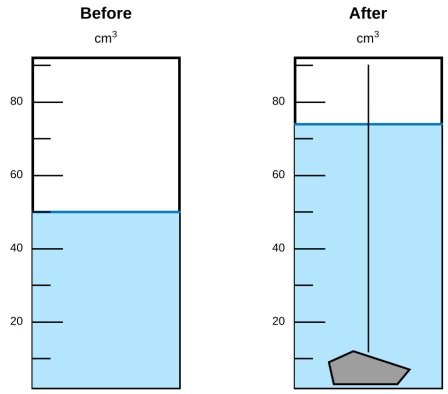
Answer = _____ kg

0 2 . 5 Ice has a density of 920 kg/m^3. Calculate the volume of 2.3 kg of ice. **[3 marks]**

Answer = _____ m^3

0 2 . 6 Explain why ice floats on water. **[2 marks]**

0 3 A student finds the density of an irregular stone. She measures the mass of the stone as 190 g. She then lowers the stone into a measuring cylinder containing water, as shown.



0 3 . 1 Use the diagrams to find the volume of the stone. **[2 marks]**

0 3 . 2 Calculate the density of the stone. **[3 marks]**

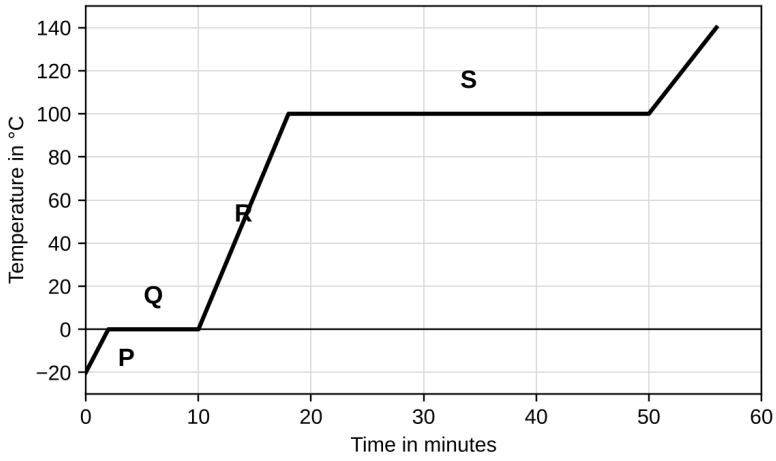
Answer = _____ g/cm^3

0 3 . 3 State **one** way the student can make sure the reading of the volume is accurate. **[1 mark]**

0 3 . 4 Describe how the student could find the density of a **regular** solid cuboid. **[4 marks]**

0 3 . 5 Suggest why the method with the measuring cylinder would not work for a piece of cork. **[2 marks]**

0 4 A solid substance is heated steadily until it becomes a gas. The graph shows how the temperature changes with time.



0 4 . 1 State the melting point and the boiling point of the substance. **[2 marks]**

0 4 . 2 Which part of the graph, **P**, **Q**, **R** or **S**, shows the substance melting? **[1 mark]**

0 4 . 3 What is happening to the substance in part **R**? **[1 mark]**

0 4 . 4 Explain why the temperature stays constant during part **S** although the substance is being heated. **[3 marks]**

0 4 . 5 Use the graph to find the time for which the substance was boiling. **[2 marks]**

0 4 . 6 Explain why part **S** is much longer than part **Q**. **[2 marks]**

0 5 This question is about internal energy.

0 5 . 1 What is meant by the **internal energy** of a substance? **[2 marks]**

0 5 . 2 Describe how internal energy changes when a liquid is heated and its temperature rises. **[2 marks]**

0 5 . 3 Explain what happens to the potential energy of the particles when a liquid boils. **[2 marks]**

0 5 . 4 A bicycle pump gets warm when it is used. Explain why. **[3 marks]**

0 6 Specific latent heat of fusion of ice = 334 000 J/kg. Specific latent heat of vaporisation of water = 2.26×10^6 J/kg. Specific heat capacity of water = 4200 J/kg °C.

0 6 . 1 What is meant by **specific latent heat**? **[2 marks]**

0 6 . 2 Write down the equation that links energy, mass and specific latent heat. **[1 mark]**

0 6 . 3 Calculate the energy needed to melt 2.0 kg of ice at 0 °C. **[3 marks]**

Answer = _____ J

0 6 . 4 Calculate the energy needed to boil away 0.50 kg of water at 100 °C. **[3 marks]**

Answer = _____ J

0 6 . 5 A 2.0 kW heater transfers all its energy to the water in part 4. Calculate the time taken. **[3 marks]**

Answer = _____ s

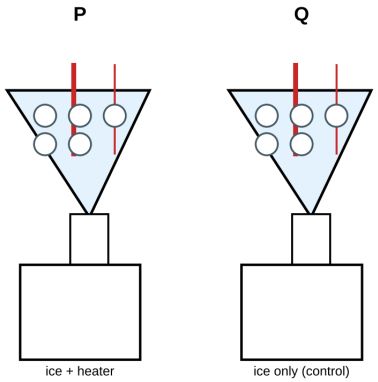
0 6 . 6 Calculate the total energy needed to heat 0.10 kg of water from 20 °C to 100 °C and then boil it all away. **[5 marks]**

Answer = _____ J

0 6 . 7 A student melts 0.30 kg of ice using 100 200 J. Calculate the specific latent heat of fusion of ice. **[3 marks]**

Answer = _____ J/kg

0 7 A student determines the specific latent heat of fusion of ice. She places crushed ice in two funnels, **P** and **Q**. Funnel **P** contains a 50 W heater. Funnel **Q** has no heater. She collects the melted water for 5.0 minutes. The mass of water collected from P is 0.062 kg and from Q is 0.017 kg.



0 7 . 1 Explain why the student sets up funnel **Q**. **[2 marks]**

0 7 . 2 Calculate the mass of ice melted **by the heater**. **[1 mark]**

0 7 . 3 Calculate the energy supplied by the heater in 5.0 minutes. **[3 marks]**

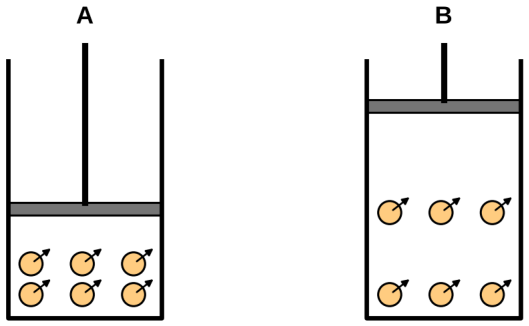
Answer = _____ J

0 7 . 4 Calculate the specific latent heat of fusion of ice from her results. **[3 marks]**

Answer = _____ J/kg

0 7 . 5 Suggest **one** way the student could improve the accuracy of her value for L. **[1 mark]**

0 8 The diagrams show a fixed mass of gas in a cylinder with a piston. The temperature of the gas is the same in **A** and **B**.



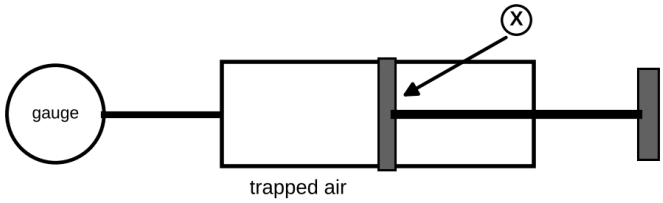
0 8 . 1 Explain how the gas particles cause pressure on the walls of the cylinder. **[3 marks]**

0 8 . 2 In which cylinder, **A** or **B**, is the pressure greater? Explain your answer. **[3 marks]**

0 8 . 3 The gas in cylinder **B** is heated and the piston is fixed. Explain what happens to the pressure. **[3 marks]**

0 8 . 4 What is the relationship between the temperature of a gas and the average kinetic energy of its particles? **[1 mark]**

0 9 A student investigates how the pressure of a fixed mass of gas depends on its volume. The apparatus is shown. The temperature is kept constant. The graph shows her results.



0 9 . 1 Name the part labelled **X** and state what it does in the investigation. **[2 marks]**

0 9 . 2 Describe the relationship between the pressure and the volume of the gas. **[2 marks]**

0 9 . 3 Use the graph to find the pressure when the volume is 60 cm^3. **[1 mark]**

0 9 . 4 The gas has a pressure of 120 kPa when its volume is 100 cm^3. It is compressed at constant temperature to a pressure of 240 kPa. Calculate the new volume. **[4 marks]**

Answer = _____ cm^3

0 9 . 5 Explain why the student compresses the gas slowly. **[2 marks]**

_____

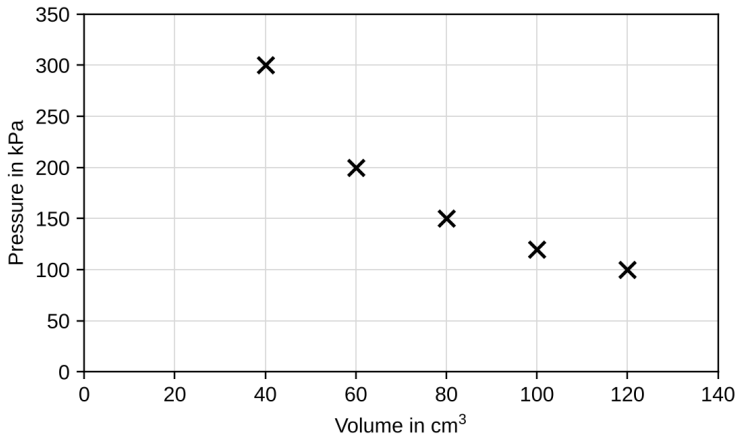
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0 9 . 6 A scuba cylinder contains air at 20 000 kPa and has a volume of 0.010 m^3. Calculate the volume this air would occupy at 100 kPa and the same temperature. **[3 marks]**

Answer = _____ m^3

1 0 The student's results for pressure and volume are plotted below.

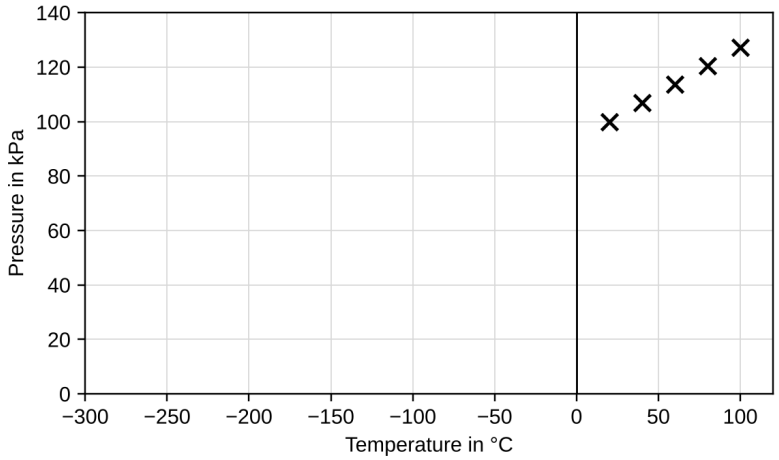


1 0 . 1 Show that pressure × volume is constant for the results at 40 cm^3 and 100 cm^3. **[3 marks]**

Answer = _____

1 0 . 2 Draw a smooth curve of best fit on the graph. **[1 mark]**

1 1 A fixed volume of gas is heated. The graph shows how its pressure changes with temperature.



1 1 . 1 Draw a straight line of best fit through the points and extend it to meet the horizontal axis. **[2 marks]**

1 1 . 2 Use your line to estimate the value of absolute zero in °C. **[2 marks]**

1 1 . 3 Convert 25 °C to kelvin. **[1 mark]**

1 1 . 4 Convert 350 K to degrees Celsius. **[1 mark]**

1 1 . 5 Explain what is meant by absolute zero and why the pressure of an ideal gas would be zero there. **[2 marks]**

1 2 This question is about evaporation and boiling.

1 2 . 1 Give **one** difference between evaporation and boiling. **[1 mark]**

1 2 . 2 Explain why a cup of tea cools faster when it is stirred or blown across. **[3 marks]**

1 2 . 3 State what is meant by **sublimation**. **[1 mark]**

12.4 A closed container of gas is placed in a freezer. State what happens to the average kinetic energy of the particles. **[1 mark]**

13 Read the question and answer in full sentences.

13.1 *Explain, using the particle model, the differences between the properties of solids, liquids and gases. Your answer should include the arrangement and movement of the particles and the density of each state. **[6 marks]**

13.2 *A student heats a beaker of ice from −20 °C until the water boils. Describe and explain how the temperature and the energy of the particles change during the experiment. **[6 marks]**

END OF QUESTIONS

MARK SCHEME – Topic 4.3: Particle model of matter

How to use this mark scheme: mark your own work in a different colour. A semicolon (;) separates marking points and each is worth one mark unless stated. A slash (/) separates alternative answers that are equally acceptable. Words in brackets are not needed to gain the mark. **Allow** = acceptable; **Ignore** = neither credited nor penalised; **Do not accept** = contradicts or spoils the answer. Marks are for the *science*, not the wording.

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.1	A: solid ; B: liquid ; C: gas ;		3	AO1 4.3.1.1
01.2	vibrate ; about fixed positions ;		2	AO1 4.3.1.1
01.3	in a gas the particles are far apart ; so there is a lot of space (empty) to move into ; in a solid the particles are already touching / close together ;		3	AO2 4.3.1.1
01.4	the particles in a gas are much further apart ; so there are fewer particles in the same volume ; so less mass per unit volume ;		3	AO2 4.3.1.1
01.5	it stays the same (is conserved)		1	AO1 4.3.2.1
01.6	the material recovers its original properties if the change is reversed ; no new substance is formed / only the arrangement of particles changes ;		2	AO2 4.3.2.1
02.1	density = mass ÷ volume		1	AO1 4.3.1.1
02.2	V = 4.0 × 3.0 × 2.0 = 24 (cm^3) ; ρ = m ÷ V ; = 64.8 ÷ 24 ; 2.7 (g/cm^3) ;		4	AO2 4.3.1.1
02.3	× 1000 ; 2700 (kg/m^3) ;	1 g/cm^3 = 1000 kg/m^3.	2	AO2 4.3.1.1
02.4	m = ρ × V ; = 1000 × 0.50 ; 500 (kg) ;		3	AO2 4.3.1.1
02.5	V = m ÷ ρ ; = 2.3 ÷ 920 ; 0.0025 (m^3) ;	Allow 2.5 × 10^{-3}.	3	AO2 4.3.1.1
02.6	ice is less dense than water ; water expands when it freezes (particles are further apart in ice) ;		2	AO2 4.3.1.1
03.1	74 − 50 ; 24 (cm^3) ;		2	AO3 4.3.1.2 RP
03.2	ρ = m ÷ V ; = 190 ÷ 24 ; 7.9 (g/cm^3) ;	Allow 7.92. Allow ecf from part 1.	3	AO2 4.3.1.2 RP

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.3	read the bottom of the meniscus at eye level ; make sure no air bubbles stick to the stone ;	Any one.	1	AO3 4.3.1.2 RP
03.4	measure the mass using a balance ; measure length, width and height with a ruler ; calculate volume = l × w × h ; calculate density = mass ÷ volume ;		4	AO1 4.3.1.2 RP
03.5	cork floats / is less dense than water ; so cannot be fully submerged (need to push it under / use a sinker) ;		2	AO3 4.3.1.2 RP
04.1	melting point 0 °C ; boiling point 100 °C ;		2	AO3 4.3.2.1
04.2	Q		1	AO3 4.3.2.1
04.3	the liquid is being heated / temperature of the liquid is rising		1	AO3 4.3.2.1
04.4	the energy supplied is used to break the bonds between particles / change the state ; not to increase the kinetic energy of the particles ; so the temperature does not rise ;		3	AO2 4.3.2.2
04.5	50 − 18 ; 32 (minutes) ;		2	AO3 WS 3.1
04.6	more energy is needed to change a liquid to a gas than a solid to a liquid ; (specific latent heat of vaporisation is greater than that of fusion / more bonds broken) ;		2	AO3 4.3.2.3
05.1	the total kinetic energy and potential energy ; of all the particles (that make up the system) ;		2	AO1 4.3.2.2
05.2	internal energy increases ; particles gain kinetic energy / move faster ;		2	AO1 4.3.2.2
05.3	potential energy increases ; particles move further apart (bonds broken) ;		2	AO2 4.3.2.2
05.4	work is done on the gas by the force on the plunger ; energy is transferred to the internal energy / kinetic energy of the gas particles ; so the temperature of the gas rises ;		3	AO2 4.3.3.2
06.1	the energy needed to change the state of 1 kg of a substance ; without changing its temperature ;		2	AO1 4.3.2.3
06.2	thermal energy for a change of state = mass × specific latent heat		1	AO1 4.3.2.3
06.3	E = m L ; = 2.0 × 334 000 ; 668 000 (J) ;		3	AO2 4.3.2.3
06.4	$0.50 \times 2.26 \times 10^6$; 1 130 000 ; J ;	Allow 1.13×10^6 J.	3	AO2 4.3.2.3

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.5	t = E ÷ P ; = 1 130 000 ÷ 2000 ; 565 (s) ;	Allow ecf from part 4.	3	AO2 4.3.2.3
06.6	ΔE = 0.10 × 4200 × 80 ; = 33 600 (J) ; E = 0.10 × 2.26 × 10^6 = 226 000 (J) ; total = 33 600 + 226 000 ; 259 600 (J) ;		5	AO3 4.3.2.3
06.7	L = E ÷ m ; = 100 200 ÷ 0.30 ; 334 000 (J/kg) ;		3	AO2 4.3.2.3
07.1	control experiment ; to find the mass of ice that melts because of the room temperature / not the heater ;		2	AO3 4.3.2.3 RP
07.2	0.062 − 0.017 = 0.045 (kg)		1	AO3 4.3.2.3 RP
07.3	t = 300 s ; E = P × t = 50 × 300 ; 15 000 (J) ;		3	AO2 4.3.2.3 RP
07.4	L = E ÷ m ; = 15 000 ÷ 0.045 ; 333 000 (J/kg) / 3.3 × 10^5 ;	Allow 330 000 to 335 000.	3	AO3 4.3.2.3 RP
07.5	repeat and calculate a mean ; run the experiment for longer so more ice melts ; insulate the funnels / use a lid ; use a heater with a measured p.d. and current ;	Any one sensible improvement.	1	AO3 4.3.2.3 RP
08.1	particles move randomly and collide with the walls ; each collision exerts a force (change of momentum) ; pressure = force on the walls per unit area ;		3	AO1 4.3.3.1
08.2	A ; particles are in a smaller volume ; so collisions with the walls are more frequent / more per second ;		3	AO2 4.3.3.2
08.3	particles gain kinetic energy / move faster ; collisions are more frequent and with more force ; so pressure increases ;		3	AO2 4.3.3.1
08.4	the higher the temperature, the greater the average kinetic energy (proportional to the temperature in kelvin)		1	AO1 4.3.3.1
09.1	plunger / piston ; changes the volume of the trapped air ;		2	AO1 4.3.3.2
09.2	as volume increases pressure decreases ; pressure is inversely proportional to volume / pressure × volume is constant ;		2	AO3 4.3.3.2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
09.3	200 (kPa)		1	AO3 WS 3.1
09.4	$p_1V_1 = p_2V_2$; 120 × 100 = 240 × V ; V = 12 000 ÷ 240 ; 50 (cm^3) ;		4	AO2 4.3.3.2
09.5	compressing quickly heats the gas ; temperature must stay constant for the relationship to hold ;		2	AO3 4.3.3.2
09.6	20 000 × 0.010 = 100 × V ; V = 200 ÷ 100 ; 2.0 (m^3) ;		3	AO2 4.3.3.2
10.1	40 × 300 = 12 000 ; 100 × 120 = 12 000 ; both equal / same product ;		3	AO3 4.3.3.2
10.2	smooth curve through all points, decreasing, not touching axes ;		1	AO3 WS 3.3
11.1	straight line of best fit ; line extrapolated back to zero pressure ;		2	AO3 WS 3.3
11.2	line meets axis at ... ; −273 °C (accept −260 to −290) ;		2	AO3 4.3.3.1
11.3	298 (K)	Allow 25 + 273.	1	AO2 4.3.3.1
11.4	77 (°C)		1	AO2 4.3.3.1
11.5	lowest possible temperature (0 K) ; particles have no kinetic energy so do not collide with the walls ;		2	AO2 4.3.3.1
12.1	evaporation happens at the surface at any temperature; boiling happens throughout the liquid at the boiling point		1	AO1 4.3.2
12.2	the particles with the most energy escape from the surface ; this removes the particles with the highest kinetic energy so the average falls ; blowing removes vapour / increases the evaporation rate ;		3	AO3 4.3.2
12.3	a solid changes directly to a gas (without becoming liquid)		1	AO1 4.3.2.1
12.4	it decreases		1	AO2 4.3.3.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
13.1	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • Level 3 (5–6 marks): clear comparison of all three states including arrangement, movement and a link to density. • Level 2 (3–4): description of at least two states with some explanation. • Level 1 (1–2): simple statements about one or two states. • Indicative content: solid – regular arrangement, particles touching, vibrate about fixed positions, high density, fixed shape and volume; liquid – irregular, particles close together, move past each other, similar density to solid, fixed volume but takes shape of container; gas – far apart, move randomly at high speed, very low density, fills container, compressible; density differences due to particle spacing in the same volume.		6	AO1 4.3.1.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
13.2	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • Level 3 (5–6): a full description of all heating stages with correct particle-level explanations of the plateaus. • Level 2 (3–4): description of some stages with some explanation. • Level 1 (1–2): simple statements. • Indicative content: temperature rises as the ice is heated (particles vibrate faster); at 0 °C temperature stays constant while melting; energy used to break bonds / increase potential energy; temperature rises as water is heated; at 100 °C temperature constant while boiling; energy used to overcome forces between particles (potential energy increases); latent heat; temperature rises for steam.		6	AO2 4.3.2

Where marks are lost in Topic 4.3

Use this page after marking. Tick the ones that cost you marks, then redo those questions closed-book.

- ☐ **Density:** ρ = m ÷ V. Check units: g/cm³ × 1000 = kg/m³. Show the equation, substitution and answer.
- ☐ **Describe the particles** in each state using three ideas: **arrangement**, **movement**, **spacing**. Gas particles are far apart, not 'lighter'.
- ☐ **During a change of state** temperature is constant. The energy goes into breaking bonds (potential energy), not increasing kinetic energy.
- ☐ **Specific latent heat:** E = m L (no temperature change). Specific heat capacity: ΔE = m c Δθ. Use the right one for each stage and add the results.
- ☐ **Mass is conserved** in a change of state; it is a physical (reversible) change.
- ☐ **Gas pressure:** pressure comes from collisions with the walls. Say **frequency** and **force** of collisions. Smaller volume → more frequent collisions.
- ☐ **Constant temperature** is required for p V = constant. Say that you compress slowly so the gas stays at the same temperature.
- ☐ **Kelvin:** K = °C + 273. Absolute zero = 0 K = −273 °C (kinetic energy is zero).